JEE Mathematics: Quick Review with Grading and Practice Map - Permutations & Combinations (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (85 questions; a question that appears in two of these is counted once): High = 8 or more questions, Medium = 1 to 7 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Fundamental Principle of Counting

Question 01

When to add and when to multiply the number of ways?

Model Answer

Steps that are all carried out one after another ("and") multiply: $m \times n$. Alternatives of which exactly one happens ("or") add: $m + n$.

Ex: 4 roads Delhi→Jaipur and 3 roads Jaipur→Bombay: Delhi to Bombay via Jaipur $4 \times 3 = 12$; ways of leaving Jaipur $4 + 3 = 7$.

Grade - Low

Practice map, main concept: DPP: 1(4-5)

Also used in: DPP: 1(2-3); 4(3)

Question 02

When to use the complementary method?

Model Answer

Use it for "at least", "not" or "neither" conditions: required = total − (cases in which the condition fails).

Ex: 3-digit numbers with at least one digit 7: $900 - 8 \cdot 9 \cdot 9 = 252$.

Grade - Medium

Practice map, main concept: Sheet: J-I 6; J-M 7, 29; C-I 17 | DPP: 1(15-17)

Also used in: Sheet: J-I 10, 19; J-II 8, 32 | DPP: 3(4, 8-9); 4(5, 17); 5(13); 6(7-8, 12, 19) | Question Bank: 192

Question 03

How to deal with "at least" and "at most" in questions?

Model Answer

At least r: add the cases $r, r+1, \ldots$ or subtract the cases below r from the total. At most r: add the cases $0, 1, \ldots, r$.

Ex: committee of 5 from 6 men and 4 women with at least 2 women: ${}^{10}C_5 - {}^{6}C_5 - {}^{4}C_1\,{}^{6}C_4 = 186$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23; J-II 32; C-I 3 | DPP: 3(7) | Question Bank: 165

Also used in: Sheet: J-I 6; J-II 2-4, 7, 11, 43, 48; J-M 2, 5; J-A 8, 12; C-II 7 | DPP: 3(3, 11); 4(13) | Question Bank: 167, 194

Question 04

How to count "neither / none / at least one of several conditions" using inclusion-exclusion (Venn diagram)?

Model Answer

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

None of the conditions = total $-n(A \cup B \cup C)$; "exactly" or "at least" counts follow from the Venn regions. Ex: integers 1 to 100 divisible by none of $2, 3, 5$: $100 - (50 + 33 + 20) + (16 + 10 + 6) - 3 = 26$.

Grade - Medium

Practice map, main concept: Sheet: J-I 19-20; J-II 31; J-M 21, 23; C-II 2 | DPP: 6(10) | Question Bank: 199

Also used in: Sheet: J-M 3 | DPP: 3(14) | Question Bank: 182

Formation of Numbers

Question 05

When should we take repetition as allowed or not allowed in the formation of numbers?

Model Answer

Digits repeat only if the question says so ("repetition allowed"); otherwise each digit is used at most once. The first place of a number cannot hold 0. Fill the most restricted place first.

Ex: 3-digit numbers: with repetition $9 \cdot 10 \cdot 10 = 900$; without repetition $9 \cdot 9 \cdot 8 = 648$.

Grade - High

Practice map, main concept: Sheet: J-I 7; J-II 5; J-M 4, 8, 19, 24; J-A 17; C-I 13; C-II 12 | DPP: 1(3, 7-10); 2(2, 4, 7, 9); 3(4, 6, 9, 15); 6(4, 18) | Question Bank: 171

Also used in: Sheet: J-M 31; J-A 11 | DPP: 1(5, 11, 17); 2(3, 17); 3(7, 16-17) | Question Bank: 184

Question 06

How to find the number of numbers divisible by 3?

Model Answer

A number is divisible by 3 iff its digit sum is. List the digit sets whose sum is a multiple of 3 (use remainders mod 3), arrange each set and subtract the arrangements that start with 0.

Ex: 5-digit numbers from $0, 1, 2, 3, 4, 5$ without repetition: sets $\{1, 2, 3, 4, 5\}$ (sum 15) and $\{0, 1, 2, 4, 5\}$ (sum 12): $120 + (120 - 24) = 216$.

Grade - Medium

Practice map, main concept: Sheet: J-II 8; C-II 1 | DPP: 1(18) | Question Bank: 183, 192

Also used in: Sheet: J-M 15 | DPP: 1(13)

Question 07

How to find the number of numbers divisible by 4?

Model Answer

A number is divisible by 4 iff its last two digits form a multiple of 4. List the allowed endings, then fill the remaining places.

Ex: 4-digit numbers from $0, 1, 2, 3, 4, 5$ without repetition divisible by 4: endings $04, 20, 40$ give $3 \times (4 \cdot 3) = 36$ and endings $12, 24, 32, 52$ give $4 \times (3 \cdot 3) = 36$ (0 cannot lead); total 72.

Grade - Medium

Practice map, main concept: Sheet: J-M 28; J-A 11 | DPP: 3(17)

Also used in: none

Question 08

How to find the numbers divisible by 2, 5, 6, 8, 9, 11 or 25?

Model Answer

Use the test: 2 last digit even; 5 last digit 0 or 5; 6 both 2 and 3; 8 last three digits; 9 digit sum; 11 (sum at odd places − sum at even places) divisible by 11; 25 last two digits $00, 25, 50, 75$.

Fix the last digit(s) first, then fill the other places. Ex: 4-digit numbers without repetition divisible by 5 (digits 0 to 9): last digit 0 gives $9 \cdot 8 \cdot 7 = 504$, last digit 5 gives $8 \cdot 8 \cdot 7 = 448$; total 952.

Grade - Medium

Practice map, main concept: Sheet: J-M 10, 15, 31 | DPP: 1(11, 13); 2(17); 4(9)

Also used in: Sheet: J-M 11, 21 | DPP: 1(7); 2(2, 9); 6(4)

Question 09

How to count (and add) palindromic numbers or words?

Model Answer

A palindrome is fixed by its first half (including the middle place): an n-letter palindrome has $\lceil n/2 \rceil$ free places; the first digit of a number is non-zero.

Ex: 3-digit palindromes $9 \cdot 10 = 90$; 7-letter palindromes from 26 letters 26^4; for the sum, add place values digit by digit.

Grade - Medium

Practice map, main concept: Sheet: J-I 29; J-M 11 | DPP: 1(12); 6(9)

Also used in: none

Question 10

How to count numbers (or selections) whose digits are strictly increasing or decreasing?

Model Answer

Choose the set of digits; its order is then fixed. Increasing n-digit numbers use digits 1 to 9: ${}^{9}C_{n}$; decreasing ones may also use 0: ${}^{10}C_{n}$. The same idea holds for numbers drawn and written in sorted order.

Ex: 5-digit numbers with digits increasing: ${}^{9}C_{5} = 126$; decreasing: ${}^{10}C_{5} = 252$; their difference is 126.

Grade - Medium

Practice map, main concept: Sheet: J-II 13; C-I 12 | DPP: 3(12) | Question Bank: 168

Also used in: Sheet: J-II 57; J-A 16

m^n or n^m

Question 11

How to solve m^n or n^m type questions?

Model Answer

Ask "who has a choice?": the objects being assigned each choose among all the places, so the answer is $(\text{number of places})^{(\text{number of objects})}$.

Ex: 6 persons in 5 compartments 5^6; 3 rings on 4 fingers 4^3; m toys among n children (any number each) n^m.

Grade - Medium

Practice map, main concept: Sheet: J-II 57; J-M 17; C-I 15; C-II 13, 15 | DPP: 1(1); 2(12, 14); 3(8); 5(11); 6(8)

Also used in: Sheet: J-M 29; C-I 17 | DPP: 2(10)

Rank of Word

Question 12

How to find the rank of a word when the letters are different?

Model Answer

Write the letters alphabetically. For each letter of the word from the left, count the smaller unused letters $\times$ (places left)!; add all and add 1.

Ex: RAPID (order A, D, I, P, R): words before it start with A, D, I, P ($4 \times 24 = 96$), then RAD___ (2), RAI___ (2), RAPD_ (1): $96 + 2 + 2 + 1 = 101$ words before it, so rank $= 102$.

Grade - Low

Practice map, main concept: DPP: 1(19, 21)

Also used in: DPP: 1(14)

Question 13

How to find the word at a given position when the formed words are arranged alphabetically?

Model Answer

Work backwards: divide (position -1) by $(n-1)!$; the quotient picks the first letter from the sorted list; continue with the remainder and $(n-2)!$, and so on (use alike-letter block sizes when letters repeat).

Ex: 50th word from A, D, I, P, R: $49 = 2 \cdot 24 + 1$, so I; then $1 = 0 \cdot 6 + 1$ gives A; $1 = 0 \cdot 2 + 1$ gives D; $1 = 1 \cdot 1 + 0$ gives R then P: IADRP.

Grade - Medium

Practice map, main concept: Sheet: J-II 10, 25 | DPP: 1(14, 20) | Question Bank: 175

Also used in: Sheet: C-I 12

Factorial

Question 14

Have you remembered the factorial values up to 7!?

Model Answer

$$0! = 1,\ 1! = 1,\ 2! = 2,\ 3! = 6,\ 4! = 24,\ 5! = 120,\ 6! = 720,\ 7! = 5040,\ 8! = 40320$$

$$n! = n\,(n-1)!, \quad n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}, \quad (2n)! = 2^n\, n!\,[1 \cdot 3 \cdot 5 \cdots (2n-1)]$$

Grade - Medium

Practice map, main concept: Sheet: J-II 15, 50; C-I 16

Also used in: Sheet: J-M 1 | Question Bank: 180

Question 15

How to find the power of a prime number in $n!$?

Model Answer

$$E_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

Ex: $E_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$; $E_3(100!) = 33 + 11 + 3 + 1 = 48$.

Grade - Low

Practice map, main concept: Sheet: J-II 17; C-II 4

Also used in: DPP: 1(22)

Question 16

How to find the power of a composite number in $n!$?

Model Answer

Factorise $N = p^a q^b \cdots$; the power of N in $n!$ is $\min\left(\left\lfloor \frac{E_p}{a} \right\rfloor, \left\lfloor \frac{E_q}{b} \right\rfloor, \ldots\right)$.

Ex: power of 6 in 50!: $E_2 = 47,\ E_3 = 22$, so $\min(47, 22) = 22$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 17

How to find the number of zeroes at the end of $n!$?

Model Answer

Zeroes = power of 10 = $E_5(n!)$, because 5 occurs less often than 2.

Ex: 100! ends in $20 + 4 = 24$ zeroes; 45! in $9 + 1 = 10$ zeroes.

Grade - Low

Practice map, main concept: Sheet: J-II 18 | DPP: 1(22)

Also used in: none

Properties of ${}^nC_r$

Question 18

Write down all 6 properties of ${}^nC_r$.

Model Answer

$$ {}^nC_r = \frac{n!}{r!\,(n-r)!}, \quad {}^nP_r = {}^nC_r \cdot r! $$

$$ {}^nC_r = {}^nC_{n-r}, \quad {}^nC_0 = {}^nC_n = 1; \qquad {}^nC_x = {}^nC_y \Rightarrow x = y \text{ or } x + y = n $$

$$ {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r, \quad {}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1}, \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} $$

Ex: ${}^nC_3 = {}^nC_5 \Rightarrow n = 3 + 5 = 8$, so ${}^nC_2 = {}^8C_2 = 28$.

Grade - Medium

Practice map, main concept: Sheet: J-I 5; J-II 16, 53; J-M 1, 30 | DPP: 4(1, 7) | Question Bank: 180

Also used in: Sheet: J-M 9; J-A 9

Questions based on Selection and Arrangements

Question 19

When to apply ${}^nP_r$ and when to apply ${}^nC_r$?

Model Answer

Only a group is chosen (committee, team, hand, subset) $\Rightarrow {}^nC_r$. Choosing followed by an order (words, numbers, positions, prizes) $\Rightarrow {}^nP_r = {}^nC_r\, r!$.

Ex: first three places among 10 swimmers: ${}^{10}P_3 = 720$; choosing any 3 of them: ${}^{10}C_3 = 120$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3; J-II 1, 9; J-A 16; C-II 14, 16 | DPP: 1(2, 6); 2(1, 5, 10, 13); 3(16); 4(2); 6(5)

Also used in: Sheet: J-I 5, 9; J-II 13; J-M 7, 17, 23, 30; C-I 3; C-II 11, 28 | DPP: 2(4, 7); 4(1); 5(9, 14) | Question Bank: 165, 198

Question 20
Do you know all the basic terms associated with playing cards?

Model Answer
A pack has 52 cards: 4 suits (spades, clubs black; hearts, diamonds red) of 13 cards each (A, 2 to 10, J, Q, K); 4 aces, 12 face cards, 26 red and 26 black cards.

Ex: 5-card hands of one suit: $4 \cdot {}^{13}C_5 = 5148$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 6

Question 21
How to find the total number of lines and triangles formed by many points, some of them collinear?

Model Answer
Lines $= {}^{n}C_2 - {}^{p}C_2 + 1$, triangles $= {}^{n}C_3 - {}^{p}C_3$ (p of the n points collinear)

For several collinear sets subtract each set's ${}^{p}C_3$. Ex: 10 points, 6 collinear: triangles $120-20 = 100$, lines $45 - 15 + 1 = 31$.

Grade - Medium
Practice map, main concept: Sheet: J-I 4 | DPP: 2(6); 3(13); 4(8) | Question Bank: 166, 174
Also used in: Sheet: J-II 56; C-II 28

Question 22
How to find the total number of quadrilaterals formed by many points, some of them collinear?

Model Answer
Quadrilaterals $= {}^{n}C_4 - {}^{p}C_4 - {}^{p}C_3 \cdot {}^{n-p}C_1$ (p collinear points)

Remove the choices in which 3 or 4 of the chosen vertices lie on the line. Ex: 12 points, 5 collinear: $495 - 5 - 10 \cdot 7 = 420$.

Grade - Low
Practice map, main concept: Sheet: C-I 18
Also used in: none

Question 23
What is the number of diagonals of an n-sided polygon?

Model Answer
Diagonals $= {}^{n}C_2 - n = \dfrac{n(n-3)}{2}$

Ex: a regular polygon with interior angle 150° has $n = 12$ sides, so $\dfrac{12 \cdot 9}{2} = 54$ diagonals.

Grade - Medium
Practice map, main concept: Sheet: J-A 6 | DPP: 3(5)
Also used in: none

Question 24
How to select three numbers in A.P.?

Model Answer
The first and third terms must be of the same parity (the middle term is then fixed): count pairs of equal parity from $1, 2, \ldots, n$.

$$n \text{ odd}: \ \frac{(n-1)^2}{4}, \qquad n \text{ even}: \ \frac{n(n-2)}{4}$$

Ex: from 1 to 10: $2 \cdot {}^{5}C_2 = 20$.

Grade - Low

Practice map, main concept: Sheet: C-I 20; C-II 9

Also used in: none

Question 25

How to find the number of rectangles and squares in a chess board (grid)?

Model Answer

Grid of $m \times n$ unit squares: rectangles $= {}^{m+1}C_2 \cdot {}^{n+1}C_2$, squares $= \sum_{k=1}^{\min(m,n)} (m-k+1)(n-k+1)$

Rectangles that are not squares = rectangles − squares. Ex: 8×8 board: 1296 rectangles, 204 squares.

Grade - Low

Practice map, main concept: Sheet: C-I 8; C-II 22-24 | DPP: 3(2); 4(5)

Also used in: Sheet: C-II 14

Question 26

How to count selections with conditions: particular persons always included, never included, together or not at all, or not together?

Model Answer

Always included: choose the rest from the remaining. Never included: choose from the others. Together or not at all: add the two cases. Two who refuse to be together: total − (both). Categories (boys / girls, bowlers / batsmen): split by cases and use at least / at most.

Ex: committee of 5 from 9 where A, B serve together or not at all and C, D never serve together: $({}^7C_3 - {}^5C_1) + ({}^7C_5 - {}^5C_3) = 30 + 11 = 41$.

Grade - High

Practice map, main concept: Sheet: J-I 8; J-II 2-4, 6-7, 11-12, 42-44, 48; J-M 2, 5, 20; J-A 8-9, 12, 18; C-I 2; C-II 7 | DPP: 3(1, 3, 10-11); 4(11, 13-14, 17) | Question Bank: 167, 170, 190, 194, 198

Also used in: Sheet: J-A 19 | DPP: 5(8)

Question 27

How to find the number of intersection points of lines, diagonals or chords?

Model Answer

n lines (no two parallel, no three concurrent): ${}^nC_2$ points; p parallel lines: $-{}^pC_2$; q concurrent lines: $-{}^qC_2 + 1$

Chords of n points on a circle (no three concurrent) meet inside in ${}^nC_4$ points. Ex: 9 points: 126; 20 lines with 10 parallel and 10 concurrent: $190 - 45 - 45 + 1 = 101$.

Grade - Medium

Practice map, main concept: Sheet: C-I 11 | Question Bank: 196

Also used in: none

Question 28
How to arrange when some persons or objects are tied to particular places (seats, middle place, sides of a table)?

Model Answer
Fill the restricted places first (choose who goes there and arrange them), then arrange the others in the remaining places.

Ex: choose 4 novels of 6 and 1 dictionary of 3 and arrange them with the dictionary in the middle: ${}^{3}C_{1} \cdot {}^{6}C_{4} \cdot 4! = 1080$.

Grade - Low
Practice map, main concept: Sheet: J-I 9; J-II 14 | DPP: 4(4, 12, 16); 5(14); 6(14)
Also used in: Sheet: J-I 15; J-II 55

Gap and String Method

Question 29
How and when to use the string method?

Model Answer
When some objects must be together: tie them into one block, arrange the blocks ($n-k+1$ units), then arrange inside the block: $(n-k+1)!\,k!$.

"Not together" = total − together. Ex: 5 boys and 3 girls in a row, not all girls together: $8!-6!\,3! = 36000$.

Grade - Medium
Practice map, main concept: Sheet: J-I 10; J-A 7 | DPP: 5(1, 3, 12); 6(12) | Question Bank: 186
Also used in: Sheet: J-II 19; J-M 13-14 | DPP: 4(4); 6(14) | Question Bank: 169, 199

Question 30
How and when to use the gap method?

Model Answer
When some objects must be apart (no two together): arrange the others first, then put the restricted objects in the gaps, at most one per gap: $(n-r)! \cdot {}^{n-r+1}P_{r}$.

Ex: 5 boys and 3 girls in a row, no two girls together: $5! \cdot {}^{6}P_{3} = 14400$.

Grade - Medium
Practice map, main concept: Sheet: J-II 19; J-M 13
Also used in: Sheet: J-A 7 | DPP: 4(16) | Question Bank: 186

Question 31
Can we use the string method when two types of groups are to be kept together?

Model Answer
Yes: make one block for each type, arrange the blocks among themselves, then arrange inside every block.

Ex: 3 girls and 4 boys in a queue with all girls together and all boys together: $2! \cdot 3! \cdot 4! = 288$.

Grade - Medium
Practice map, main concept: DPP: 6(3) | Question Bank: 169
Also used in: none

Question 32
How to arrange when two types of things are to be kept alternate?

Model Answer
Equal numbers n, n: $2 \cdot n!\, n!$ (either type first). Numbers $n+1$ and n: the larger type must occupy both ends, $(n+1)!\, n!$. A difference of 2 or more is impossible.

Ex: 4 boys and 4 girls alternately: $2 \cdot 4! \cdot 4! = 1152$; 5 boys and 4 girls: $5! \cdot 4! = 2880$.

Grade - Low
Practice map, main concept: none
Also used in: none

Station Method

Question 33
When to use the station method?

Model Answer
When r objects are to be chosen from n objects in a row so that no two chosen ones are adjacent (stops of a train, persons or points in a line).

Ex: a train with 12 stations stops at 5 of them, no two consecutive: ${}^{8}C_{5} = 56$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 34
How to use the station method?

Model Answer

$$^{n-r+1}C_{r}$$

Put the $n-r$ unchosen objects in a row; they make $n-r+1$ stations (gaps); choose r stations.
Ex: 4 of 10 points in a row, no two consecutive: ${}^{7}C_{4} = 35$.

Grade - Medium
Practice map, main concept: Sheet: J-A 14 | DPP: 5(4-5, 13); 6(15)
Also used in: none

Formation of Groups

Question 35
How to find the number of ways of forming groups of distinct things into given group sizes?

Model Answer
Unequal sizes p, q, r: $\dfrac{n!}{p!\, q!\, r!}$; $\quad$ k unlabelled groups of equal size p: $\dfrac{(kp)!}{(p!)^{k}\, k!}$

If equal groups are given to different persons (labelled) do not divide by $k!$. Ex: 10 persons into groups of $2, 3, 5$: 2520; into groups of $4, 3, 3$: $\dfrac{10!}{4!\, 3!\, 3!\, 2!} = 2100$.

Grade - Medium
Practice map, main concept: Sheet: J-I 12-13; J-II 20-21, 23, 49; J-A 19; C-I 1, 6, 19; C-II 19-21 | DPP: 5(6, 8, 15) | Question Bank: 176, 178
Also used in: Sheet: J-II 24; J-A 15; C-II 18 | DPP: 5(7, 10)

Question 36
When to use formation of groups in questions?

Model Answer
When distinct things are divided or distributed into groups, teams, cars or persons with given sizes. If the sizes are not fixed, list every size pattern, count each by the group formula, and multiply by the ways to hand the groups to the persons.

Ex: 8 persons in 3 cars of different make, at most 3 in a car: pattern $(3,3,2)$: $3 \cdot \dfrac{8!}{3!\,3!\,2!} = 1680$.

Grade - Medium
Practice map, main concept: Sheet: J-II 22, 24; J-A 15 | DPP: 5(7, 10)
Also used in: Sheet: J-I 12; J-II 21; C-II 20

Question 37
How to distribute n things among 2 or 3 persons when each person gets at least one thing?

Model Answer
Two persons: $2^n - 2$; three persons: $3^n - 3 \cdot 2^n + 3$

Total − (one person empty) + (two persons empty). Ex: 5 different balls among 3 persons: $243 - 96 + 3 = 150$.

Grade - Medium
Practice map, main concept: Sheet: J-A 1
Also used in: Sheet: J-II 51 | DPP: 5(15)

Permutation of Alike Objects taken all at a time

Question 38
How to arrange things when a few of them are alike?

Model Answer

$$\frac{n!}{p!\,q!\,r!\cdots} \qquad (p, q, r \text{ alike objects of each kind})$$

Ex: words from MATHEMATICS (M, A, T twice each) starting and ending with T: $\dfrac{9!}{2!\,2!} = 90720$.

Grade - Medium
Practice map, main concept: Sheet: J-I 14-15; J-II 26, 55; J-A 10 | DPP: 4(3); 5(9) | Question Bank: 189
Also used in: Sheet: J-II 22, 31, 52; J-M 19, 27; C-II 8 | DPP: 4(7); 5(2, 12); 6(3, 9, 18)

Question 39
How to use the string method in permutation of alike objects taken all at a time?

Model Answer
Tie the objects that must be together into one block; count the arrangements of the block's own letters (alike formula) and of the units outside the block (alike formula); multiply.

Ex: INDEPENDENCE with all vowels (I, E, E, E, E) together: $\dfrac{5!}{4!} \cdot \dfrac{8!}{3!\,2!} = 16800$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 55

Question 40

How to use the gap method in permutation of alike objects taken all at a time?

Model Answer

Arrange the other objects (alike formula), then choose the gaps for the restricted objects: ${}^{m+1}C_r$ if they are alike (times $r!$ if they are different).

Ex: MISSISSIPPI with no two S together: $\frac{7!}{4!\,2!} \cdot {}^{8}C_4 = 105 \cdot 70 = 7350$.

Grade - Medium

Practice map, main concept: Sheet: J-I 16; J-II 27; J-M 22 | DPP: 5(2); 6(1)

Also used in: Sheet: J-II 53, 57

Question 41

What does "without changing the order of vowels / consonants" mean in questions?

Model Answer

Their relative order is fixed: arrange all objects freely and divide by $k!$ (or choose their k places in ${}^{n}C_k$ ways and fill the rest). "Positions of vowels fixed" means the vowels do not move at all.

Ex: MULTIPLE (L twice, vowels U, I, E) without changing the order of vowels, other ways: $\frac{8!}{2!\,3!} - 1 = 3359$.

Grade - Low

Practice map, main concept: Sheet: J-II 28 | DPP: 2(8, 16); 3(14); 4(6, 10, 15); 6(17)

Also used in: DPP: 5(1)

Question 42

How to arrange things when two or three types of things are to be kept separated?

Model Answer

For a kind that must not be adjacent use the gap method; for two or three such kinds use inclusion-exclusion: total $-$ (some kind has a pair together) $+ \cdots$, tying each pair into a block.

Ex: HONOLULU (O, L, U twice) with no two alike letters together: 2220.

Grade - Low

Practice map, main concept: Sheet: J-II 46-47 | DPP: 6(7)

Also used in: none

Question 43

How to find the rank of a word when letters are alike?

Model Answer

As for different letters, but the number of words starting with each smaller letter is (remaining letters)! divided by the factorials of the alike letters still left.

Ex: SMALL (order A, L, L, M, S): A: 12, L: 24, M: 12, SA: 3, SL: 6, so 57 words come before it and the rank is 58.

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 17; J-M 14, 26; C-I 4 | Question Bank: 193

Also used in: Sheet: J-II 25

Permutation of Alike Objects taken some at a time

Question 44

How to find the number of selections when alike things are taken some at a time?

Model Answer

Split into types by the pattern of repeats (e.g. "two alike", "all different") and count the selections of each type; or use the coefficient method.

Ex: 2 letters from DADDY (D three times, A, Y): DD 1, two different 3; total 4.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-II 30, 45, 54; C-II 6 | DPP: 6(2) | Question Bank: 177, 197

Question 45

How to find the number of arrangements when alike things are taken some at a time?

Model Answer

For every type of selection multiply its number of selections by the arrangements $\frac{r!}{(\text{alike})!}$ of that type; then add the types.

Ex: arrangements of 2 letters from DADDY: $1 + 3 \cdot 2! = 7$.

Grade - Medium

Practice map, main concept: Sheet: J-II 29-30; C-I 10; C-II 8 | DPP: 6(2) | Question Bank: 197

Also used in: Sheet: J-A 10

Circular Permutations

Question 46

What is circular permutation and how is it different from linear permutation?

Model Answer

In a circle only relative positions matter (a rotation gives the same arrangement), so there is no first place: linear $n!$, circular $(n-1)!$. If the seats are numbered or distinguishable, rotation gives a new arrangement and the answer is $n!$.

Ex: 4 persons round a table $3! = 6$; on numbered chairs $4! = 24$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 47

How to arrange things in circular permutation?

Model Answer

All n different things: $(n-1)!$; r of n different things: $\frac{{}^{n}P_{r}}{r}$

Fix one object and arrange the rest. Ex: choose 3 of 5 persons and seat them round a table: $\frac{{}^{5}P_{3}}{3} = 20$.

Grade - Medium

Practice map, main concept: DPP: 6(11, 16)

Also used in: Sheet: J-I 11; J-A 13

Question 48
How to use the string method in circular permutation?

Model Answer

$$(n-k)! \cdot k! \qquad (k \text{ objects together in a circle of } n)$$

Tie the k objects into one block ($n-k+1$ units give $(n-k)!$ circular ways), then arrange inside. Ex: 10 persons round a table with P_1, P_2, P_3 together: $7! \cdot 3!$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 21-22; J-II 33 | DPP: 6(19-20)
Also used in: none

Question 49
How to use the gap method in circular permutation?

Model Answer

$$(m-1)! \cdot {}^{m}P_{r} \qquad (m \text{ unrestricted objects, } r \text{ restricted ones kept apart})$$

Seat the m unrestricted ones in a circle; they leave m gaps (not $m+1$). Ex: 6 men and 5 women round a table, no two women together: $5! \cdot {}^{6}P_{5} = 6!\,5!$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 11
Also used in: none

Question 50
When will we consider clockwise and anti-clockwise arrangements as same, and what to do in that case?

Model Answer
When the arrangement can be turned over (garland, necklace, bracelet, beads strung on a thread): divide the circular count by 2, $\dfrac{(n-1)!}{2}$ for $n \geq 3$.

Ex: 8 different flowers strung in a garland with 4 particular flowers together: $\dfrac{4! \cdot 4!}{2} = 288$.

Grade - **Low**
Practice map, main concept: none
Also used in: Sheet: J-I 21-22 | DPP: 6(11)

Question 51
When will we consider clockwise and anti-clockwise arrangements as same in case of arrangement of persons?

Model Answer
Persons at a table: the two orders are different (each person's left and right neighbours swap), so do not divide by 2. Divide only when the question says the arrangements are alike if every person has the same two neighbours.

Ex: 5 persons at a round table $4! = 24$ (beads on a necklace: 12).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 52
How to use permutation of alike objects taken all at a time in circular permutations?

Model Answer
Fix one object that occurs only once (or a distinct one) to remove the rotation, then arrange the rest in a line with the alike formula.

Ex: one flag A, three alike B and two alike C round a circle: fix A, then $\frac{5!}{3!\,2!} = 10$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 53

Station Method in Circular Permutation

Question 53
When to use the station method in circular permutations?

Model Answer
When r of n objects placed on a circle (points, vertices, persons at a round table) are chosen so that no two chosen are adjacent.

Ex: choosing 3 of 9 persons round a table so that no two are neighbours: $\frac{9}{6} \cdot {}^{6}C_3 = 30$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 54
How to solve formation-of-polygon type questions in circular permutations?

Model Answer

$$\frac{n}{n-r}\,{}^{n-r}C_r \qquad (r \text{ vertices of an } n\text{-gon, no side common})$$

Ex: triangles from the vertices of a regular octagon with no side of the octagon: $\frac{8}{5} \cdot {}^{5}C_3 = 16$; for a nonagon 30.

Grade - Low
Practice map, main concept: DPP: 6(13)
Also used in: DPP: 6(15)

Question 55
How to use the station method in circular permutations? (both methods)

Model Answer
Method 1: ${}^{n-r}C_r + {}^{n-r-1}C_{r-1}$ (first object not chosen / chosen); Method 2: $\frac{n}{n-r}\,{}^{n-r}C_r$

Method 1 cuts the circle at one object and uses the row formula in two cases. Ex: $n = 8, r = 3$: $10 + 6 = 16 = \frac{8}{5} \cdot 10$.

Grade - Low
Practice map, main concept: none
Also used in: none

Total Number of Combinations

Question 56
How to find the total number of combinations when many things are given, some alike of one type, some alike of another type, etc.?

Model Answer

$(p+1)(q+1)(r+1)\cdot 2^k$ $(p, q, r$ alike of three kinds, k different things; empty selection included)

Ex: one or more balls from 10 white, 9 green and 7 black identical balls: $11 \cdot 10 \cdot 8 - 1 = 879$.

Grade - Low
Practice map, main concept: Sheet: J-I 24; J-II 54, 56 | DPP: 7(1)
Also used in: none

Question 57
When to subtract 1 from the answer in total-number-of-combinations questions?

Model Answer
The product counts the empty selection once; subtract 1 when "one or more" (at least one) must be chosen, not when "none" is allowed.

Ex: one or more of 10 questions, each with an alternative (3 choices each: skip, question, alternative): $3^{10} - 1$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 24 | DPP: 7(1)

Divisors Based Problems

Question 58
How to find the total number of divisors?

Model Answer

$$N = p^a q^b r^c \Rightarrow d(N) = (a+1)(b+1)(c+1)$$

Ex: $21600 = 2^5\, 3^3\, 5^2$ has $6 \cdot 4 \cdot 3 = 72$ divisors.

Grade - Medium
Practice map, main concept: Sheet: J-II 35; C-II 10 | Question Bank: 187
Also used in: Sheet: J-I 25; J-II 34; C-I 7

Question 59
How to find the total number of odd and even divisors?

Model Answer
Odd divisors $= (b+1)(c+1)$ (omit the prime 2); even divisors $= a(b+1)(c+1)$

Ex: 21600: odd $4 \cdot 3 = 12$, even $5 \cdot 4 \cdot 3 = 60$.

Grade - Medium
Practice map, main concept: Sheet: J-M 6; C-I 7
Also used in: none

Question 60
What do you mean by proper divisors?

Model Answer
Proper divisors are the divisors other than 1 and N itself: $d(N) - 2$.

Ex: 21600 has $72 - 2 = 70$ proper divisors.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 61
How to find the total number of divisors divisible by a given number?

Model Answer
If the divisor must be a multiple of $M = p^{a'}q^{b'}\cdots$, take each exponent from its minimum: $(a-a'+1)(b-b'+1)\cdots$. "Not divisible by K" = all $-$ divisible by K.

Ex: $1980 = 2^2\,3^2\,5\cdot 11$: 36 divisors, of which multiples of 11: $3\cdot 3\cdot 2\cdot 1 = 18$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 25; J-II 34
Also used in: none

Question 62
How to find the sum of all divisors?

Model Answer

$$\sigma(N) = \frac{p^{a+1}-1}{p-1}\cdot\frac{q^{b+1}-1}{q-1}\cdot\frac{r^{c+1}-1}{r-1}$$

Ex: 21600: $63\cdot 40\cdot 31 = 78120$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 26
Also used in: Sheet: J-II 35

Question 63
How to find the sum of all odd and even divisors?

Model Answer
Odd divisors: drop the factor of the prime 2, $(1+3+\cdots+3^b)(1+5+\cdots+5^c)$. Even divisors: $\sigma(N)-$ (odd sum).

Ex: 21600: odd sum $40\cdot 31 = 1240$, even sum $78120 - 1240 = 76880$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 64
How to find the sum of all divisors divisible by a given number?

Model Answer
Divisors that are multiples of M are $M\times$ (divisors of N/M): sum $= M\,\sigma(N/M)$.

Ex: divisors of 21600 divisible by 10: $10\,\sigma(2160) = 10\cdot 7440 = 74400$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 65

How to find the number of ways in which a given number can be expressed as a product of two divisors?

Model Answer

$$\frac{d(N)}{2} \ (N \text{ not a perfect square}), \qquad \frac{d(N)+1}{2} \ (N \text{ a perfect square})$$

Unordered pairs $N = a \cdot b$; ordered pairs are $d(N)$. Ex: $7056 = 2^4\,3^2\,7^2$ is a square, $d = 45$: $\frac{45+1}{2} = 23$.

Grade - Low

Practice map, main concept: Sheet: J-II 36

Also used in: Sheet: J-II 35; C-II 5

Question 66

How to find the number of ways in which a given number can be expressed as a product of two divisors which are relatively prime?

Model Answer

$$2^{k-1} \text{ (unordered)}, \qquad 2^k \text{ (ordered)}, \qquad k = \text{number of distinct primes of } N$$

Each prime power goes wholly to one of the two factors. Ex: $300300 = 2^2\,3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$, $k = 6$: $2^5 = 32$.

Grade - Low

Practice map, main concept: Sheet: J-I 27

Also used in: Sheet: J-II 35

Question 67

How to find the product of all divisors?

Model Answer

$$\prod \text{divisors} = N^{\,d(N)/2}$$

Divisors pair up with product N. Ex: $N = 12$: $1 \cdot 2 \cdot 3 \cdot 4 \cdot 6 \cdot 12 = 12^3 = 1728$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 68

How to count ordered pairs of numbers with a given LCM or GCD?

Model Answer

LCM $= p^a q^b \cdots$: for each prime the larger exponent must be a, giving $2a+1$ pairs of exponents, so ordered pairs $= \prod(2a_i + 1)$. GCD $= g$: write $x = ga$, $y = gb$ with $\gcd(a, b) = 1$ and count coprime pairs in the allowed range.

Ex: LCM $= r^2t^4s^2$: $5 \cdot 9 \cdot 5 = 225$ ordered pairs.

Grade - Medium

Practice map, main concept: Sheet: C-I 9 | Question Bank: 185

Also used in: none

Maximum value of ${}^nC_r$

Question 69

How to find the maximum value of ${}^nC_r$?

Model Answer

${}^nC_r$ is greatest at $r = \dfrac{n}{2}$ (n even); $\qquad r = \dfrac{n-1}{2}$ or $\dfrac{n+1}{2}$ (n odd)

Ex: ${}^{40}C_r$ is greatest at $r = 20$; ${}^{37}C_r$ at $r = 18$ or 19.

Grade - Low

Practice map, main concept: Sheet: J-II 52

Also used in: Sheet: J-II 54

Grid Problem

Question 70

How to find the number of ways in a grid structure?

Model Answer

Shortest path $(0,0) \to (a,b)$: $\dfrac{(a+b)!}{a!\,b!} = {}^{a+b}C_a$

Arrange a horizontal and b vertical steps. Ex: $(0,0) \to (9,8)$: ${}^{17}C_9 = 24310$.

Grade - Low

Practice map, main concept: Sheet: J-I 28

Also used in: none

Question 71

How to find the number of ways in a grid structure if we have to pass through a given point every time?

Model Answer

Multiply the paths of the two parts. Paths that avoid a point = total − paths through it.

Ex: $(0,0) \to (9,9)$ avoiding $(5,5)$: ${}^{18}C_9 - {}^{10}C_5\,{}^{8}C_4 = 48620 - 17640 = 30980$.

Grade - Low

Practice map, main concept: Sheet: J-II 37

Also used in: none

Question 72

How to find the number of ways in a grid structure if exactly r diagonal steps are to be taken?

Model Answer

$$\frac{(a+b-r)!}{r!\,(a-r)!\,(b-r)!} \qquad (r \text{ diagonal},\ a-r \text{ horizontal},\ b-r \text{ vertical steps})$$

Ex: $(2,1) \to (11,12)$ with exactly one diagonal step: $\dfrac{19!}{8!\,10!\,1!}$.

Grade - Low

Practice map, main concept: none

Also used in: none

De-arrangement

Question 73

What is the subjective approach to find D_n?

Model Answer

Choose which r objects stay in their own places and derange the other $n-r$: $n! = \sum_{r=0}^{n} {}^{n}C_r \, D_{n-r}$, with $D_0 = 1,\ D_1 = 0$; solve for D_n step by step.

Ex: $24 = D_4 + 4D_3 + 6D_2 + 4D_1 + D_0 = D_4 + 8 + 6 + 0 + 1$, so $D_4 = 9$.

Grade - Medium

Practice map, main concept: Sheet: J-A 4

Also used in: none

Question 74

What is the formula for D_n?

Model Answer

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right), \qquad D_n = (n-1)(D_{n-1} + D_{n-2})$$

$D_1 = 0,\ D_2 = 1,\ D_3 = 2,\ D_4 = 9,\ D_5 = 44,\ D_6 = 265$. Ex: 5 letters, at least 3 wrongly placed: $D_5 + {}^{5}C_1 D_4 + {}^{5}C_2 D_3 = 44 + 45 + 20 = 109$.

Grade - Medium

Practice map, main concept: Sheet: J-II 40; C-II 25-27

Also used in: Sheet: J-A 4

Summation of Numbers

Question 75

How to use the place value method in finding the sum of numbers formed by given digits?

Model Answer

Sum $= f \times$ (sum of the digits) $\times \underbrace{11\cdots1}_{\text{number of places}}$ (f = times a digit occupies one place)

If 0 is among the digits, find the sum with 0 allowed in front and subtract the sum of the numbers that begin with 0. Ex: all 4-digit numbers from $1, 3, 5, 7$: $3! \cdot 16 \cdot 1111 = 106656$.

Grade - Low

Practice map, main concept: Sheet: J-I 30-31; C-I 14

Also used in: none

Question 76

How to use the carry method in finding the sum of numbers formed by given digits?

Model Answer

Add column by column: column sum $= f \times$ (sum of digits); write its last digit and carry the rest to the next column.

Ex: numbers from $1, 2, 3$ (no repetition): each column sums $2 \cdot 6 = 12$; $12 \cdot 111 = 1332$.

Grade - Low

Practice map, main concept: none

Also used in: none

Distribution of Alike Objects

Question 77

When to use the beggars' method?

Model Answer

When identical things are distributed among distinct persons (boxes, children) and each may get none, one or more.

Ex: 30 coins among 5 persons; choosing 6 chocolates from 8 brands.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 78

How to use the beggars' method?

Model Answer

n alike things among p persons (none allowed): ${}^{n+p-1}C_{p-1}$

Ex: 30 coins among 5 persons: ${}^{34}C_4$; 6 chocolates from 8 brands: ${}^{13}C_7 = {}^{13}C_6$.

Grade - Low

Practice map, main concept: Sheet: J-I 32; C-II 11

Also used in: Sheet: J-I 34

Question 79

How to use the beggars' method when there is a minimum condition from the beggars' side?

Model Answer

Person i gets at least m_i: ${}^{n-\sum m_i+p-1}C_{p-1}$; each at least one: ${}^{n-1}C_{p-1}$

Give the minimum first, then distribute the rest freely. Ex: 20 oranges among 3 children, each at least one: ${}^{19}C_2 = 171$; 30 coins, $A \geq 3, B \geq 2, C \geq 3$ among 5: ${}^{26}C_4$.

Grade - Medium

Practice map, main concept: Sheet: J-I 34; C-II 28

Also used in: Sheet: J-II 53; J-M 22

Algebra based problems in distribution of alike objects

Question 80

How to find the number of non-negative integral solutions of an equation of type $x + y + z = 100$?

Model Answer

$x_1 + x_2 + \cdots + x_k = n$ has ${}^{n+k-1}C_{k-1}$ non-negative and ${}^{n-1}C_{k-1}$ positive (natural) solutions

Ex: $x + y + z = 100$: ${}^{102}C_2$ non-negative, ${}^{99}C_2$ natural.

Grade - Medium

Practice map, main concept: Question Bank: 182

Also used in: Sheet: J-I 18; J-II 38 | DPP: 6(6)

Question 81

How to find the number of non-negative integral solutions of an equation of type $x + y + z = 100$ when conditions on x, y, z are given?

Model Answer

Lower bounds: put $x = a + x'$ and so on (a negative lower bound works the same way). Upper bounds: subtract the solutions that break the bound, or use the coefficient method.

Ex: $x \geq 3,\ y \geq 2,\ z \geq 1$ with $x + y + z = 100$: ${}^{96}C_2$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 38; J-M 16 | DPP: 6(6)

Also used in: Sheet: C-I 5 | Question Bank: 172, 188, 191

Question 82

How to find the number of non-negative integral solutions of an inequality such as $x+y+z+w \leq 30$?

Model Answer

$$x_1 + \cdots + x_k \leq n \ \Leftrightarrow\ x_1 + \cdots + x_k + p = n,\ p \geq 0 \text{ (dummy variable)} : \ {}^{n+k}C_k$$

Strict inequality $< n$ means $\leq n - 1$. Ex: $x + y + z + w \leq 30$: ${}^{34}C_4$; $x + y + z + w < 30$: ${}^{33}C_4$.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 83

How to find the number of non-negative integral solutions of an inequality when conditions on x, y, z are given?

Model Answer

First shift every variable by its minimum (reducing n), then add the dummy variable.

Ex: natural solutions of $x + y + z + w \leq 30$: put $x = 1 + x'$ etc. to get $x' + y' + z' + w' \leq 26$: ${}^{30}C_4$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 35

Also used in: Sheet: J-I 36

Question 84

How to find the number of positive integral solutions of equations of type $xyz = 30$?

Model Answer

$N = p^a q^b r \Rightarrow$ ordered solutions of $x_1 x_2 \cdots x_k = N$: ${}^{a+k-1}C_{k-1} \cdot {}^{b+k-1}C_{k-1} \cdots$

Distribute the exponent of each prime among the k variables. Ex: $xyz = 30$: $3 \cdot 3 \cdot 3 = 27$; $xyz = 21600$: ${}^{7}C_2 \cdot {}^{5}C_2 \cdot {}^{4}C_2 = 1260$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 33; J-II 39; J-M 27; C-II 3, 5

Also used in: none

Coefficient Method

Question 85

When to use the coefficient method?

Model Answer

When each variable or kind of object is bounded above (dice totals, digit sums, selections from alike objects, equal sharing): the count is a coefficient in a product of polynomials.

Ex: dividing 6 identical green, 6 blue and 6 red things equally between two persons is the coefficient of x^9 in $(1 + x + \cdots + x^6)^3$, which is 37.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 86

How to write the factors in the coefficient method and which power of x is to be computed?

Model Answer

A variable with values a to b gives the factor $x^a + x^{a+1} + \cdots + x^b$; answer = coefficient of x^n (n = the total) in the product

Take out x^a and use $\dfrac{1 - x^{m+1}}{1 - x}$ with the series of $(1 - x)^{-k}$. Ex: three dice with total 11: coefficient of x^8 in $(1 + x + \cdots + x^5)^3 = {}^{10}C_2 - 3 \cdot {}^4C_2 = 27$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18, 36; J-II 41; J-M 18; C-I 5 | Question Bank: 172, 188, 191

Also used in: Sheet: J-M 16; J-A 18

Question 87

How to use the coefficient method in finding the number of selections in permutation of alike objects taken some at a time?

Model Answer

A kind with p alike objects gives $(1 + x + \cdots + x^p)$, each different object gives $(1 + x)$; the number of selections of r objects is the coefficient of x^r.

Ex: 5 letters from MISSISSIPPI: coefficient of x^5 in $(1 + x)(1 + x + \cdots + x^4)^2(1 + x + x^2)$ is 25.

Grade - Medium

Practice map, main concept: Sheet: J-II 45; C-II 6 | Question Bank: 177

Also used in: none

Miscellaneous

Question 88

How to express the general term of a series involving factorial as a difference of two terms?

Model Answer

$$r \cdot r! = (r + 1)! - r!, \qquad \frac{r}{(r + 1)!} = \frac{1}{r!} - \frac{1}{(r + 1)!}$$

Write T_r as $f(r + 1) - f(r)$ so that the sum telescopes. Ex: $\sum_{r=1}^{n} r \cdot r! = (n + 1)! - 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 1; J-M 9, 25 | Question Bank: 173

Also used in: none

Question 89

How to count by remainder classes (parity or residues mod m) when a sum must be divisible by m?

Model Answer

Sort the elements by remainder mod m; list the combinations of remainders whose sum is $\equiv 0$; multiply the class sizes. For parity the last choice is made by the class that fixes the parity (half of all).

Ex: ordered pairs of distinct $x, y \in \{1, \ldots, 25\}$ with $x + y$ divisible by 5 (5 numbers in each class): $2\,({}^5C_2 + 5 \cdot 5 + 5 \cdot 5) = 120$.

Grade - Medium

Practice map, main concept: Sheet: J-M 12 | Question Bank: 184

Also used in: Sheet: J-M 6

Question 90

How to count arrangements in which adjacent objects must differ (colouring a line or a circle)?

Model Answer

n places in a line, k colours, neighbours different: $k(k-1)^{n-1}$; in a circle: $(k-1)^n + (-1)^n(k-1)$

The first place has k choices and each next place $k-1$ (a circle needs the closing condition). Ex: flag of 6 strips in 4 colours: $4 \cdot 3^5 = 972$; 5 persons in a circle, 3 hat colours: $2^5 - 2 = 30$.

Grade - Medium

Practice map, main concept: Sheet: J-A 13 | DPP: 2(3, 11, 15)

Also used in: none

Extra Reading Material

Question 91

Write down the formula for the distribution of n distinct objects among k persons if each person receives at least one object.

Model Answer

$$\sum_{r=0}^{k-1} (-1)^r \, {}^kC_r \, (k-r)^n \;=\; k^n - {}^kC_1 (k-1)^n + {}^kC_2 (k-2)^n - \cdots$$

Ex: $n = 5,\ k = 3$: $243 - 3 \cdot 32 + 3 \cdot 1 = 150$.

Grade - Medium

Practice map, main concept: Sheet: J-II 51

Also used in: Sheet: J-A 1; C-II 2

Question 92

Write down the recurrence relation of the Fibonacci sequence and write down the Fibonacci sequence also.

Model Answer

$$F_n = F_{n-1} + F_{n-2}, \quad F_1 = F_2 = 1: \quad 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

Counting sequences with no two consecutive 0's (strings of $0, 1$) obey it. Ex: 3-digit strings with no two consecutive 0's: $5 = F_5$.

Grade - Medium

Practice map, main concept: Sheet: J-A 2-3

Also used in: none

Question 93
What does a partition of a natural number mean?

Model Answer
A partition of n is a way of writing n as a sum of positive integers, the order of the parts being ignored; $p(n)$ denotes their number.
Ex: $p(4) = 5$: $4,\ 3+1,\ 2+2,\ 2+1+1,\ 1+1+1+1$; $p(8) = 22$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 94
How to write the recurrence relation of $P(n)$ and how to find $P(n)$ with its help?

Model Answer

$$p(n) = p(n-1)+p(n-2)-p(n-5)-p(n-7)+p(n-12)+p(n-15)-\cdots, \qquad p(0) = 1,\ p(\text{negative}) = 0$$

The subtracted numbers $1, 2, 5, 7, 12, 15, \ldots$ are the generalised pentagonal numbers $\frac{k(3k\mp1)}{2}$ (signs $+, +, -, -, +, +, \ldots$). Ex: $p(8) = p(7) + p(6) - p(3) - p(1) = 15 + 11 - 3 - 1 = 22$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 95
What does the partition of a natural number into at most k parts, i.e. $P(n, k)$, mean?

Model Answer
$P(n, k)$ = number of partitions of n into at most k parts $= P(n, k-1) + P(n-k, k)$
Either at most $k-1$ parts, or exactly k parts (subtract 1 from every part). Ex: $P(8, 3) = 10$, $P(8, 4) = 15$.

Grade - Medium
Practice map, main concept: Question Bank: 179
Also used in: Sheet: J-A 5

Question 96
How to find $P(n, 1)$, $P(n, 2)$ and $P(n, 3)$?

Model Answer

$$P(n,1) = 1, \qquad P(n,2) = \left\lfloor \frac{n}{2} \right\rfloor + 1, \qquad P(n,3) = \text{nearest integer to } \frac{(n+3)^2}{12}$$

Ex: $P(8, 2) = 5$, $P(8, 3) = \dfrac{121}{12} \approx 10$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 97
How to find $P(n,k)$ if $k > 3$?

Model Answer
Use $P(n,k) = P(n,k-1) + P(n-k,k)$ and build the table from the small cases (or list the partitions). Strictly increasing $n_1 < \cdots < n_k$ with sum S: put $n_i = m_i + i - 1$ to get a partition of $S - \frac{k(k-1)}{2}$ into at most k parts with $m_i \geq 1$.
Ex: $P(10,4) = 23$; $n_1 < \cdots < n_5$ positive with sum 20: partitions of 10 into exactly 5 parts $= p(5) = 7$.

Grade - Medium
Practice map, main concept: Sheet: J-A 5
Also used in: none

Question 98
What does Euler's totient function represent and how to find it?

Model Answer

$$\varphi(N) = N\left(1 - \frac{1}{p}\right)\left(1 - \frac{1}{q}\right)\cdots, \qquad \text{sum of integers} \leq N \text{ coprime to } N = \frac{N\varphi(N)}{2}$$

$\varphi(N)$ counts the integers from 1 to N coprime to N (for $N > 1$ they pair up as x and $N - x$). Ex: $\varphi(12) = 4$ (1, 5, 7, 11), sum $= 24$.

Grade - Medium
Practice map, main concept: Sheet: J-M 3
Also used in: none

Question 99
What does the Bell number B_n represent?

Model Answer
B_n = number of ways to partition a set of n distinct objects into non-empty unlabelled subsets
$B_0 = B_1 = 1,\ B_2 = 2,\ B_3 = 5,\ B_4 = 15,\ B_5 = 52,\ B_6 = 203.$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 100
How to find B_n with the help of the triangle scheme?

Model Answer
Bell triangle: start with 1; each row begins with the last entry of the previous row; every next entry = (entry to its left) + (entry above that one). B_n is the first entry of row n (row 0 is 1).

$$1;\ 1,2;\ 2,3,5;\ 5,7,10,15;\ 15,20,27,37,52;\ \cdots \Rightarrow B_n = 1,1,2,5,15,52,\ldots$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 101
What does the Stirling subset number $S(n,k)$ represent and how to find it?

Model Answer
$S(n,k)$ = ways to split n distinct objects into k non-empty unlabelled groups; $S(n,k) = k\,S(n-1,k) + S(n-1,k-1)$

$S(n,1) = S(n,n) = 1$, $S(n,2) = 2^{n-1} - 1$, $S(n,k) = \frac{1}{k!}\sum_r (-1)^{rk} C_r (k-r)^n$. Ex: $S(5,2) = 15$, $S(5,3) = 25$.

Grade - Medium
Practice map, main concept: Sheet: C-II 18
Also used in: Question Bank: 181

Question 102
What is the relation between B_n and $S(n,k)$?

Model Answer

$$B_n = \sum_{k=1}^{n} S(n,k)$$

Ex: $B_4 = S(4,1) + S(4,2) + S(4,3) + S(4,4) = 1 + 7 + 6 + 1 = 15$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 103
How to find the number of ways to distribute n distinct or alike objects into k distinct or alike boxes?

Model Answer
distinct $\to$ distinct: k^n; alike $\to$ distinct: ${}^{n+k-1}C_{k-1}$; distinct $\to$ alike: $\sum_{j\le k} S(n,j)$; alike $\to$ alike: $P(n,k)$

Boxes may stay empty. Ex: 8 alike books in 4 alike shelves: $P(8,4) = 15$; 5 different employees in 4 alike offices: $1 + 15 + 25 + 10 = 51$.

Grade - Medium
Practice map, main concept: Question Bank: 181
Also used in: Question Bank: 179

Question 104
How to find the number of ways to distribute n distinct or alike objects into k distinct or alike boxes if each box receives at least one object?

Model Answer
distinct $\to$ distinct: $k!\,S(n,k)$; alike $\to$ distinct: ${}^{n-1}C_{k-1}$; distinct $\to$ alike: $S(n,k)$; alike $\to$ alike: $P(n-k,k)$

Ex: 8 alike things in 3 alike boxes, none empty: $P(5,3) = 5$; 5 different things in 3 alike boxes, none empty: $S(5,3) = 25$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 105

How to find the number of non-negative integral solutions of the equation $x_1 + x_2 + \cdots + x_k = n$?

Model Answer

${}^{n+k-1}C_{k-1}$ ordered solutions; with the order $0 \le x_1 \le x_2 \le \cdots \le x_k$ the count is the number of partitions of n into at most k parts, $P(n, k)$

Ex: $x_1 + x_2 + x_3 = 8$: ${}^{10}C_2 = 45$ ordered, $P(8, 3) = 10$ with $x_1 \le x_2 \le x_3$.

Grade - Low

Practice map, main concept: none

Also used in: none

Ordered and weighted solutions $(0 < x_1 \le x_2 \le \cdots \le x_k)$

Question 106

How to find the number of non-negative integral solutions of the equation $x_1+2x_2+3x_3+\cdots+kx_k = n$?

Model Answer

Number of solutions = number of partitions of n into parts of size at most $k = P(n, k)$

Here x_i is the number of parts equal to i. Ex: $x + 2y + 3z = 42$: $P(42, 3) = \dfrac{45^2}{12} \approx 169$ solutions.

Grade - Medium

Practice map, main concept: Question Bank: 195

Also used in: none